

Mary Williams

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EDUCATION

The University of Texas at Austin, Austin, TX

May 2027

- BSA Computer Science and Minor in Business
- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Software Engineering, Modern Web Applications, Computer Architecture, Professional Communications

TECHNICAL SKILLS

Languages & Frameworks: Python, C, Java, Javascript, React.js, React Native, Bootstrap

Systems & Tools: Linux, Docker, Git, GDB, Bash, Firebase

Platforms & Software: AWS, Figma, Jira, Excel

SOFTWARE PROJECTS

Personal Website – React.js, Vite, Bootstrap, Firebase, Figma

August 2025

- Constructed a responsive portfolio website from designing UI/UX in Figma and translating into responsive React components, ensuring cross-device compatibility
- Leveraged Firebase Analytics to monitor engagement metrics (page views, session duration, user flow) and optimize design for better usability
- Automated CI/CD pipeline with GitHub Actions for seamless deployment to Firebase, ensuring consistent updates in ~50% less time

AWS Serverless Slack Bot - Python, API, AWS Lambda

July 2025

- Developed an automated Slack bot to fetch NASA's Astronomy Picture of the Day via NASA's REST API and post it with description and high-res images to Slack, providing interactive daily knowledge
- Built serverless architecture using AWS Lambda to schedule daily posts with AWS EventBridge, ensuring zero-maintenance automation and reducing manual effort by 100%

Pintos Operating System - C, GDB, Git

Jan 2025 - Apr 2025

- Implemented core OS components including thread scheduling, system call handling, file systems, and virtual memory management, building a fully functional OS in C
- Utilized Linux and GDB to resolve complex kernel-level issues, improving stability and performance of the OS
- Strengthened expertise in data structures and low-level problem solving, resulting in a deeper understanding of how modern operating systems manage processes, memory, and files

Huffman Encoding File Compressor - Java, Git

Apr 2024

- Engineered a file compression tool in Java using Huffman coding, constructing optimal binary trees from character frequency counts to generate encodings that reduce file size by ~30%
- Constructed robust encoding/decoding functionality with pseudo-EOF markers and header-based tree reconstruction, ensuring reliable, lossless decompression across different file types

WORK EXPERIENCE

Vorago Technologies, Austin, Tx

May 2025 - Aug 2025

Application Engineer Intern

- Reduced manual processing time by 30% by developing a Python script to automate writing of log files
- Enhanced existing automation scripts to streamline programming and testing of client Evaluation Kits (EVKs), improving workflow efficiency
- Assembled, programmed, and tested 150 EVKs for clients, ensuring high quality and timely fulfillment

LEADERSHIP & INVOLVEMENT

Society for Advancing Gender Equity in STEM (SAGES), Mentor

Aug 2024 - Present

Women in Computer Science (WiCS), Mentor/ Member

Aug 2023 - Present

HackTx Hackathon, Participant

Oct 2024

WiCS Annual Hackathon, Participant

Feb 2025